

Dev Patel

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EDUCATION

The University of North Carolina at Chapel Hill - Chapel Hill, NC

Expected Graduation Year: 2030

Prospective Major: B.S. in Biomedical Engineering

Relevant Courses: Introductory Calculus-Based Mechanics and Relativity, Multivariable Calculus, Writing in Health and Medicine

EXPERIENCES

Independent Research Review

June '24 - Aug '25

"Gene Therapy Strategies to Combat Alcoholism"

- Evaluated the potential for gene editing mechanisms, such as CRISPR-Cas9, to optimize genes associated with ethanol metabolism in order to reduce the risk of developing alcoholism.
- Learned to synthesize complex studies, present empirical evidence, and develop a causal argument.

Researcher at the Summer Science Program hosted at Lehigh University

June - July 2025

Bacterial Genomics Cohort

- 1 of 588 selected from over 3,000 applicants to conduct bacterial genomics research.
- Revealed the predictability of resistance mechanisms, such as the optimization of efflux pumps, that *Vibrio natriegen* bacteria utilize to combat the antibiotic tetracycline.
- Crafted a manuscript and presented work at a poster symposium in front of Lehigh faculty and students.

Climate Action Pathways for Schools: Modesto City Schools District (MCS)

October 2024 - May 2026

- Collaborated with the MCS Sustainability Committee to enhance sustainability efforts across 34 school sites.
 - Analyzed energy consumption data for 34 school sites to determine the highest consuming site; crafted recommendations based on this data: applying the site's existing pool cover to save \$62,500 per year, lowering the site's pool temperature by 3 degrees to save \$15,500 per year.
 - Presented recommendations to the MCS district board, which are currently being experimented with.
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LEADERSHIP AND COMMUNITY INVOLVEMENT

Jag Closet Initiative

Dec 2024 - March 2025

Founder of a Formal Clothing Closet to Assist Underprivileged Students

- Collected over 300+ donations of formal clothing through partnerships with local businesses and organizations.
- Distributed clothing to 90+ students in need across 7 different high schools, allowing them to gain access to memorable high school events & opportunities to pursue their academic ambitions.

Student Government at Joseph A. Gregori High School

Aug 2022 - May 2026

Freshman Class President, Sophomore Class President, ASB Vice President, and ASB President

- Organized and hosted 9 monthly "student-senate" meetings each year, where 250+ students voiced opinions and concerns regarding school events and campus safety; expressed these concerns as my school's representative at monthly district board meetings each year.
 - Established a new school tradition in a Homecoming parade that celebrates 20+ clubs and unifies the school community, with 800 attendees per year.
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SKILLS

Laboratory: Gel Electrophoresis, PCR Primer Design, Chemostat Calibration, Bacterial Culturing, Bacterial Transformation, Genome Assembly, Mutational Analysis, DNA Extraction, DNA Purification

Professional/Interdisciplinary: Public Speaking, Project & Resource Management, Community Networking, Establishing Business Partnerships, Troubleshooting, Delegation, Collaboration